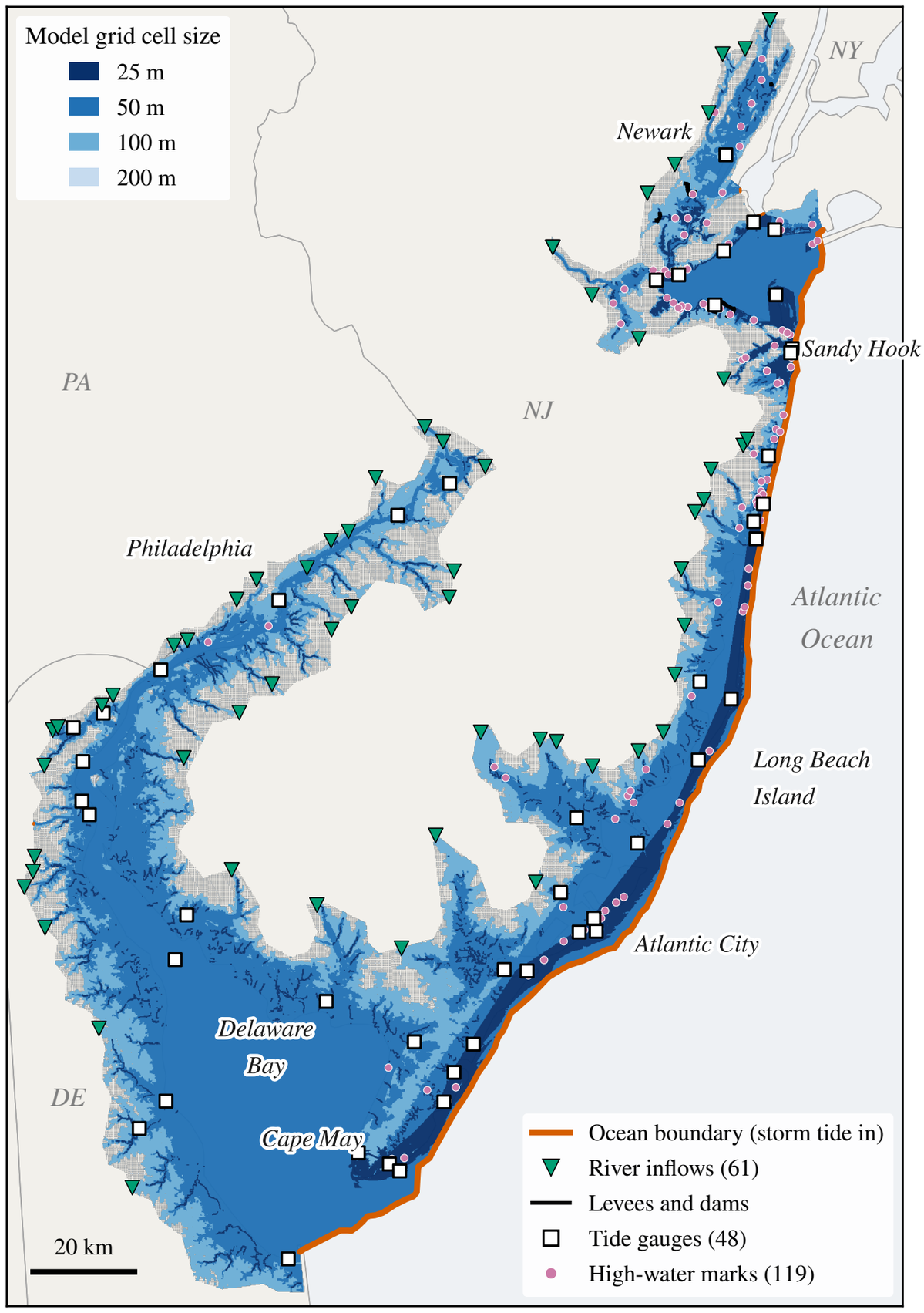


A compound-flood model of Hurricane Sandy

SFINCS, 28-31 Oct 2012 (72 h), Cape May to the Narrows, Delaware Bay and River, Raritan River and Newark Bay

3.97 million active cells
25 m cells along shores and channels
flood map on a 3.125 m ground model

Everything acting together:

Storm tide	surge + tide at the ocean boundary, USACE NACCS Sandy simulation
Waves	SnapWave, incl. long (infragravity) waves; NOAA CORA offshore
Wind, pressure	NACCS winds blended into ERA5
Rain	NOAA AORC hourly; soil soaks up part of it
Rivers	61 inflows, USGS discharge
Buildings	NJ footprints block the flow
Levees, dams	Keansburg, Elizabeth, Rahway (USACE levee database)

Checked against:

119 USGS high-water marks,
48 NOAA / USGS tide gauges,
FEMA's mapped Sandy flood extent